

B1 Fluids — Model Solutions, Trinity Term 2022

Question 1 — Darcy flow

Problem. Stack of n rigid sheets, gap $b \ll w, L$, fluid (ρ, η) , pressure drop Δp across length L . Derive velocity field and Darcy permeability Π_0 ; comment on inertia. Then generalise, prove $\nabla \cdot (\rho \mathbf{q}) = 0$, solve a spherical injection, compare two aquifers, and discuss gravity.

(a)

Take $\hat{\mathbf{x}}$ along the plates, $z \in [0, b]$ across the gap. Symmetry and no-penetration:

$$\mathbf{u} = u_x(z)\hat{\mathbf{x}}, \quad u_y = u_z = 0.$$

Steady, unidirectional, z -independent pressure: Navier–Stokes reduces to:

$$0 = -\frac{\partial p}{\partial x} + \eta \frac{\partial^2 u_x}{\partial z^2}, \quad \frac{\partial p}{\partial z} = 0.$$

Constant pressure gradient:

$$\frac{\partial p}{\partial x} = -\frac{\Delta p}{L}.$$

Integrate once in z :

$$\eta \frac{\partial u_x}{\partial z} = -\frac{\Delta p}{L} z + C_1.$$

Integrate again:

$$u_x(z) = -\frac{\Delta p}{2\eta L} z^2 + \frac{C_1}{\eta} z + C_2.$$

Apply no-slip $u_x(0) = 0$:

$$C_2 = 0.$$

Apply no-slip $u_x(b) = 0$:

$$0 = -\frac{\Delta p b^2}{2\eta L} + \frac{C_1 b}{\eta}.$$

Solve for C_1 :

$$C_1 = \frac{\Delta p b}{2L}.$$

Substitute back:

$$u_x(z) = -\frac{\Delta p}{2\eta L} z^2 + \frac{\Delta p b}{2\eta L} z.$$

Factor:

$$u_x(z) = \frac{\Delta p}{2\eta L} z(b - z).$$

Volume flux per unit cross-sectional area:

$$q = \frac{1}{b} \int_0^b u_x \, dz.$$

Substitute the velocity profile:

$$q = \frac{\Delta p}{2\eta L b} \int_0^b z(b-z) \, dz.$$

Expand the integrand:

$$\int_0^b z(b-z) \, dz = \int_0^b (bz - z^2) \, dz.$$

Evaluate term by term:

$$\int_0^b z(b-z) \, dz = \frac{b^3}{2} - \frac{b^3}{3} = \frac{b^3}{6}.$$

Substitute back:

$$q = \frac{\Delta p}{2\eta L b} \cdot \frac{b^3}{6}.$$

Simplify:

$$q = \frac{b^2}{12\eta} \frac{\Delta p}{L}.$$

Compare with $q = (\Pi_0/\eta)(\Delta p/L)$:

$$\boxed{\Pi_0 = \frac{b^2}{12}.$$

Inertia. For unidirectional $u_x(z)$: $(\mathbf{u} \cdot \nabla)\mathbf{u} = u_x \partial_x u_x \hat{\mathbf{x}} = 0$ identically. Force balance is pressure $\leftrightarrow$ viscous stress only; inertia plays no role. Equivalently $\text{Re} = Ub/\nu \ll 1$ in the Darcy regime.

(b)

Mass conservation in fixed volume V :

$$\frac{d}{dt} \int_V \rho \, dV = - \oint_{\partial V} \rho \mathbf{q} \cdot d\mathbf{S}.$$

Steady, rigid matrix: LHS=0. Apply divergence theorem:

$$\int_V \nabla \cdot (\rho \mathbf{q}) \, dV = 0.$$

Volume arbitrary, integrand continuous:

$$\boxed{\nabla \cdot (\rho \mathbf{q}) = 0.}$$

(c)

Spherical symmetry $\rho = \text{const}$ gives $\nabla \cdot \mathbf{q} = 0$ with $\mathbf{q} = q_r(r) \hat{\mathbf{r}}$:

$$\frac{1}{r^2} \frac{d}{dr} (r^2 q_r) = 0.$$

Integrate once:

$$r^2 q_r = C.$$

Solve for q_r :

$$q_r = \frac{C}{r^2}.$$

Darcy without gravity: $\mathbf{q} = -(\Pi_1/\eta)\nabla p$, so $q_r = -(\Pi_1/\eta) dp/dr$:

$$\frac{dp}{dr} = -\frac{\eta q_r}{\Pi_1} = -\frac{\eta C}{\Pi_1 r^2}.$$

Integrate from r to ∞ with $p(\infty) = p_0$:

$$\int_r^\infty \frac{dp}{dr'} dr' = -\frac{\eta C}{\Pi_1} \int_r^\infty \frac{dr'}{r'^2}.$$

Evaluate both sides:

$$p_0 - p(r) = -\frac{\eta C}{\Pi_1 r}.$$

Rearrange:

$$p(r) - p_0 = \frac{\eta C}{\Pi_1 r}.$$

Apply $p(a) = p_0 + \Delta p$:

$$\Delta p = \frac{\eta C}{\Pi_1 a}.$$

Solve for C :

$$C = \frac{\Pi_1 \Delta p a}{\eta}.$$

Substitute into q_r :

$$q_r(r) = \frac{\Pi_1 \Delta p a}{\eta r^2}.$$

Aquifer choice. Total volumetric flow rate through sphere of radius a :

$$Q = 4\pi a^2 q_r(a).$$

Substitute $q_r(a) = \Pi_1 \Delta p / (\eta a)$:

$$Q = 4\pi a^2 \cdot \frac{\Pi_1 \Delta p}{\eta a}.$$

Simplify:

$$Q = \frac{4\pi a \Pi_1 \Delta p}{\eta}.$$

Invert for Δp :

$$\Delta p = \frac{Q \eta}{4\pi a \Pi_1}.$$

Pumping power = $Q \Delta p$ (rate of work against pressure):

$$\mathcal{P} = Q \cdot \frac{Q \eta}{4\pi a \Pi_1}.$$

Simplify:

$$\mathcal{P} = \frac{Q^2 \eta}{4\pi a \Pi_1}.$$

Carman–Kozeny scaling (cf. part (a) where $\Pi_0 = b^2/12$):

$$\Pi_1 \propto d^2.$$

Use $d_A = 2d_B$:

$$\Pi_{1,A} = 4\Pi_{1,B}.$$

At fixed Q and a , ratio reduces to $(\eta/\Pi_1)_A/(\eta/\Pi_1)_B$:

$$\frac{\mathcal{P}_A}{\mathcal{P}_B} = \frac{\eta_A/\Pi_{1,A}}{\eta_B/\Pi_{1,B}}.$$

Substitute $\eta_A = 2\eta_B$ and $\Pi_{1,A} = 4\Pi_{1,B}$:

$$\frac{\mathcal{P}_A}{\mathcal{P}_B} = \frac{2\eta_B/(4\Pi_{1,B})}{\eta_B/\Pi_{1,B}}.$$

Cancel $\eta_B/\Pi_{1,B}$:

$$\frac{\mathcal{P}_A}{\mathcal{P}_B} = \frac{1}{2}.$$

Aquifer A: half the pumping power at fixed Q .

Assumptions: same source radius a , Darcy regime ($\text{Re} \ll 1$), $\Pi \propto d^2$, equal densities (gravity drops out).

(d)

Reinstate gravity, define piezometric pressure:

$$P \equiv p + \rho g z.$$

Darcy with gravity:

$$\frac{\eta}{\Pi_1} \mathbf{q} = -\nabla p - \rho g \hat{\mathbf{z}} = -\nabla P.$$

Far-field hydrostatic: $P_\infty = p_0(0)$ constant. Source: $P(a) \approx p_0(0) + \Delta p$ (for a small). The problem in P is identical to part (c):

$$q_r(r) = \frac{\Pi_1 \Delta p a}{\eta r^2} \text{ unchanged.}$$

For matched densities gravity only relabels the absolute pressure. Density mismatch $\Delta\rho$ would break spherical symmetry (buoyant plume).

Question 2 — Stagnation-point boundary layer

Problem. Inviscid stagnation flow $\psi = -Axy$ above wall $y = 0$, $A > 0$, gravity $-g\hat{\mathbf{y}}$. Distinguish Eulerian/Lagrangian acceleration; verify wall BC; find p_∞ ; derive boundary-layer (BL) equations; reduce via $\psi = -Ax\delta f(\eta)$; estimate δ and turbulence onset.

(a)

Eulerian acceleration: $\partial\mathbf{u}/\partial t$ at fixed point (river-bank probe). Lagrangian acceleration: $D\mathbf{u}/Dt = \partial_t\mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u}$ following a parcel (drifting leaf). Steady river: $\partial_t\mathbf{u} = 0$ but parcel still accelerates through rapids via $(\mathbf{u} \cdot \nabla)\mathbf{u}$. Newton's law governs the Lagrangian acceleration.

(b)

Velocity from stream function (Hiemenz convention $v_x = -\partial_y \psi$, $v_y = \partial_x \psi$, giving inflow toward wall):

$$v_x = -\frac{\partial \psi}{\partial y} = Ax, \quad v_y = \frac{\partial \psi}{\partial x} = -Ay.$$

Inviscid wall BC is no-penetration only:

$$v_y(x, 0) = 0 \quad \checkmark.$$

Steady irrotational flow: Bernoulli $p + \frac{1}{2}\rho|\mathbf{v}|^2 + \rho gy = \text{const.}$ Compute $|\mathbf{v}|^2$:

$$|\mathbf{v}|^2 = (Ax)^2 + (-Ay)^2.$$

Expand:

$$|\mathbf{v}|^2 = A^2(x^2 + y^2).$$

Fix constant via $p(0, 0) = p_0$:

$$p + \frac{1}{2}\rho A^2(x^2 + y^2) + \rho gy = p_0.$$

Solve for p :

$$p_\infty(x, y) = p_0 - \frac{1}{2}\rho A^2(x^2 + y^2) - \rho gy.$$

Streamwise gradient:

$$\frac{\partial p_\infty}{\partial x} = -\rho A^2 x.$$

(c)

At finite Re the no-slip condition $v_x(x, 0) = 0$ replaces inviscid free-slip. Outer flow gives $v_x(x, 0) = Ax \neq 0$: incompatible. Viscosity reasserts itself in a thin layer of thickness $\delta(x) \ll L$.

Scales: $L \sim x$, $\delta \ll L$. Incompressibility:

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0.$$

Order-of-magnitude:

$$\frac{V_x}{L} \sim \frac{V_y}{\delta}.$$

Solve for V_y :

$$V_y \sim \frac{\delta}{L} V_x \ll V_x.$$

x -momentum (steady, incompressible):

$$\mathbf{v} \cdot \nabla v_x = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} \right).$$

Compare viscous terms: $\nu V_x/L^2$ vs. $\nu V_x/\delta^2$:

$$\frac{\partial^2 v_x}{\partial x^2} \ll \frac{\partial^2 v_x}{\partial y^2}.$$

Convective $\sim$ dominant viscous balance:

$$\frac{V_x^2}{L} \sim \frac{\nu V_x}{\delta^2}.$$

Solve for δ :

$$\delta \sim \sqrt{\frac{\nu L}{V_x}} = L \text{Re}^{-1/2} \ll L.$$

y -momentum: each term $\lesssim (\delta/L)V_x^2/L$; at leading order:

$$\frac{\partial p}{\partial y} \approx 0.$$

Pressure across layer matches outer:

$$\frac{\partial p}{\partial x} = \frac{\partial p_\infty}{\partial x}, \quad \frac{\partial p}{\partial y} = \frac{\partial p_\infty}{\partial y}.$$

$$\boxed{\mathbf{v} \cdot \nabla v_x = \nu \frac{\partial^2 v_x}{\partial y^2} - \frac{1}{\rho} \frac{\partial p_\infty}{\partial x}, \quad \frac{\partial p}{\partial y} = \frac{\partial p_\infty}{\partial y}, \quad \nabla \cdot \mathbf{v} = 0.}$$

(d)

Try $\psi = -Ax\delta f(\eta)$, $\eta = y/\delta$, $\delta = \text{const}$:

$$v_x = -\frac{\partial \psi}{\partial y} = Ax f'(\eta), \quad v_y = \frac{\partial \psi}{\partial x} = -A\delta f(\eta).$$

Compute derivatives needed:

$$\frac{\partial v_x}{\partial x} = Af', \quad \frac{\partial v_x}{\partial y} = \frac{Ax}{\delta} f'', \quad \frac{\partial^2 v_x}{\partial y^2} = \frac{Ax}{\delta^2} f'''.$$

Convective term:

$$\mathbf{v} \cdot \nabla v_x = v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y}.$$

Substitute the four pieces:

$$\mathbf{v} \cdot \nabla v_x = (Ax f')(Af') + (-A\delta f) \left(\frac{Ax}{\delta} f'' \right).$$

Cancel δ in the second term:

$$\mathbf{v} \cdot \nabla v_x = A^2 x (f')^2 - A^2 x f f''.$$

Factor $A^2 x$:

$$\mathbf{v} \cdot \nabla v_x = A^2 x [(f')^2 - f f''].$$

Viscous term:

$$\nu \frac{\partial^2 v_x}{\partial y^2} = \frac{\nu Ax}{\delta^2} f'''.$$

Pressure gradient from part (b):

$$-\frac{1}{\rho} \frac{\partial p_\infty}{\partial x} = A^2 x.$$

Substitute into BL equation:

$$A^2 x [(f')^2 - f f''] = \frac{\nu Ax}{\delta^2} f''' + A^2 x.$$

Divide through by A^2x :

$$(f')^2 - ff'' = \frac{\nu}{A\delta^2} f''' + 1.$$

Choose δ so $\nu/(A\delta^2) = 1$:

$$\delta = \sqrt{\frac{\nu}{A}}.$$

Substitute back:

$$(f')^2 - ff'' = f''' + 1.$$

Rearrange:

$$f''' + ff'' - (f')^2 = -1.$$

Boundary conditions (no-penetration, no-slip, far-field match $f' \rightarrow 1$ so $v_x \rightarrow Ax$):

$$f(0) = 0, \quad f'(0) = 0, \quad f'(\eta) \rightarrow 1 \text{ as } \eta \rightarrow \infty.$$

(e)

From part (d):

$$\delta = \sqrt{\nu/A} \text{ (independent of } x, y, \rho, p_0).$$

Controlled by viscous diffusion rate ν vs. strain A ; the linear acceleration $-Ax$ exactly compensates the longer transit time at large x , giving a uniform-thickness layer.

Turbulence onset. Local Reynolds number on length x :

$$\text{Re}_x = \frac{|v_x| \delta}{\nu}.$$

Substitute $|v_x| = Ax$, $\delta = \sqrt{\nu/A}$:

$$\text{Re}_x = \frac{Ax\sqrt{\nu/A}}{\nu} = x\sqrt{\frac{A}{\nu}}.$$

Set transition $\text{Re}_x \sim 10^3$:

$$l \sim 10^3 \sqrt{\nu/A}.$$

Question 3 — Rayleigh–Taylor in a cylindrical tank

Problem. Closed cylinder $0 < r < a$, $-h < z < h$. Freshwater (ρ_1) below, saltwater ($\rho_2 > \rho_1$) above, interface $z = \eta(r, t)$, no surface tension, $\mathbf{g} = -g\hat{\mathbf{z}}$, no diffusion. Membrane bursts at $t = 0$. (a) Derive Laplace + linearised BCs; (b) normal modes; (c) baroclinic vorticity equation and physical mechanism.

(a)

Initial vorticity zero in each layer; uniform ρ_i , no viscosity, no baroclinic source. Kelvin's theorem: vorticity stays zero, flow irrotational:

$$\mathbf{v}_i = \nabla\phi_i.$$

Incompressibility:

$$\nabla^2\phi_i = 0 \quad (i = 1, 2).$$

Boundary conditions. Impermeable walls:

$$\left. \frac{\partial \phi_1}{\partial z} \right|_{z=-h} = 0, \quad \left. \frac{\partial \phi_2}{\partial z} \right|_{z=h} = 0, \quad \left. \frac{\partial \phi_i}{\partial r} \right|_{r=a} = 0.$$

Kinematic at $z = \eta$ from $D(z - \eta)/Dt = 0$:

$$\frac{\partial \eta}{\partial t} + v_r \frac{\partial \eta}{\partial r} = v_z.$$

Linearise: drop the $v_r \partial_r \eta$ product (small $\times$ small) and evaluate at $z = 0$:

$$\boxed{\frac{\partial \eta}{\partial t} = \left. \frac{\partial \phi_i}{\partial z} \right|_{z=0} \quad (i = 1, 2).}$$

Linearised Bernoulli (drop $|\nabla \phi|^2/2$) in each layer:

$$p_i = -\rho_i \frac{\partial \phi_i}{\partial t} - \rho_i g z + \text{const.}$$

Continuity of pressure $p_1 = p_2$ at $z = \eta$:

$$-\rho_1 \frac{\partial \phi_1}{\partial t} - \rho_1 g \eta = -\rho_2 \frac{\partial \phi_2}{\partial t} - \rho_2 g \eta.$$

Linearise (evaluate $\partial_t \phi_i$ at $z = 0$):

$$\boxed{\rho_2 \frac{\partial \phi_2}{\partial t} - \rho_1 \frac{\partial \phi_1}{\partial t} = (\rho_1 - \rho_2) g \eta \quad \text{at } z = 0.}$$

(b)

Modal ansatz consistent with bounded axisymmetric Laplacian solution:

$$\phi_i = J_0(\alpha r) [A_i e^{\alpha z} + B_i e^{-\alpha z}] e^{\sigma t}, \quad \eta = \eta_0 J_0(\alpha r) e^{\sigma t}.$$

Radial derivative using $J'_0 = -J_1$:

$$\frac{\partial \phi_i}{\partial r} = -\alpha J_1(\alpha r) [A_i e^{\alpha z} + B_i e^{-\alpha z}] e^{\sigma t}.$$

Apply impermeability $\partial_r \phi_i|_{r=a} = 0$:

$$J_1(\alpha a) = 0.$$

Hence:

$$\boxed{\alpha_n a = j_{1,n}, \quad n = 1, 2, 3, \dots}$$

Bottom wall $\partial_z \phi_1|_{-h} = 0$ forces a combination vanishing-derivative at $z = -h$:

$$\phi_1 = C_1 \cosh[\alpha_n(z + h)] J_0(\alpha_n r) e^{\sigma t}.$$

Top wall $\partial_z \phi_2|_h = 0$ analogously gives:

$$\phi_2 = C_2 \cosh[\alpha_n(z - h)] J_0(\alpha_n r) e^{\sigma t}.$$

Kinematic at $z = 0$ for layer 1:

$$\sigma\eta_0 = \alpha_n C_1 \sinh(\alpha_n h).$$

Kinematic at $z = 0$ for layer 2:

$$\sigma\eta_0 = -\alpha_n C_2 \sinh(\alpha_n h).$$

Solve for amplitudes:

$$C_1 = \frac{\sigma\eta_0}{\alpha_n \sinh(\alpha_n h)}, \quad C_2 = -\frac{\sigma\eta_0}{\alpha_n \sinh(\alpha_n h)}.$$

Insert into dynamic condition at $z = 0$ (using $\cosh(-x) = \cosh x$):

$$\rho_2 \sigma C_2 \cosh(\alpha_n h) - \rho_1 \sigma C_1 \cosh(\alpha_n h) = (\rho_1 - \rho_2) g \eta_0.$$

Substitute C_1 and C_2 :

$$\sigma \cosh(\alpha_n h) \left[-\rho_2 \frac{\sigma\eta_0}{\alpha_n \sinh(\alpha_n h)} - \rho_1 \frac{\sigma\eta_0}{\alpha_n \sinh(\alpha_n h)} \right] = (\rho_1 - \rho_2) g \eta_0.$$

Collect:

$$-\frac{\sigma^2 \eta_0 (\rho_1 + \rho_2) \cosh(\alpha_n h)}{\alpha_n \sinh(\alpha_n h)} = (\rho_1 - \rho_2) g \eta_0.$$

Divide by η_0 and use $\coth = \cosh / \sinh$:

$$\sigma^2 (\rho_1 + \rho_2) \coth(\alpha_n h) = (\rho_2 - \rho_1) g \alpha_n.$$

Solve for σ_n^2 :

$$\sigma_n^2 = \left(\frac{\rho_2 - \rho_1}{\rho_2 + \rho_1} \right) g \alpha_n \tanh(\alpha_n h).$$

Stability. $\rho_2 > \rho_1 \Rightarrow \sigma_n^2 > 0 \Rightarrow \sigma_n$ real, one root positive: *unconditionally unstable*. Limits:

$$\alpha_n h \ll 1 : \quad \sigma_n^2 \approx \frac{\rho_2 - \rho_1}{\rho_2 + \rho_1} g \alpha_n^2 h,$$

$$\alpha_n h \gg 1 : \quad \sigma_n^2 \approx \frac{\rho_2 - \rho_1}{\rho_2 + \rho_1} g \alpha_n.$$

Growth rate increases monotonically with α_n ; without surface tension or viscosity the highest geometrically allowed mode dominates.

(c)

Variable-density momentum equation:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \frac{\mathbf{F}}{\rho}.$$

Take curl, use 2D identity $\nabla \times [(\mathbf{v} \cdot \nabla) \mathbf{v}] = (\mathbf{v} \cdot \nabla) \omega$ for $\nabla \cdot \mathbf{v} = 0$:

$$\frac{\partial \omega}{\partial t} + \mathbf{v} \cdot \nabla \omega = \nabla \times \left(\frac{\mathbf{F}}{\rho} \right).$$

Inviscid: $\mathbf{F} = -\nabla p + \rho \mathbf{g}$. Body-force curl:

$$\nabla \times \mathbf{g} = 0.$$

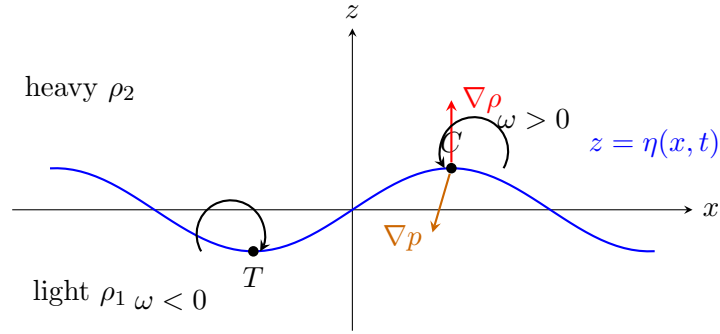
Pressure-gradient curl:

$$\nabla \times \left(-\frac{\nabla p}{\rho} \right) = -\nabla \left(\frac{1}{\rho} \right) \times \nabla p.$$

Use $\nabla(1/\rho) = -\nabla\rho/\rho^2$:

$$\boxed{\frac{\partial \omega}{\partial t} + \mathbf{v} \cdot \nabla \omega = \frac{\nabla \rho \times \nabla p}{\rho^2}.}$$

Mechanism (sketch below). Within the diffuse interface of width δ : $\nabla\rho$ is large and points from light fluid (below) into heavy fluid (above), i.e. approximately $+\hat{\mathbf{z}}$; $\nabla p \approx -\rho g \hat{\mathbf{z}}$ in the bulk but tilts horizontally where the interface is perturbed. The cross product $\nabla\rho \times \nabla p$ alternates sign along the wave: positive vorticity on one flank of each crest, negative on the other. The induced circulation lifts light fluid into crests and pushes heavy fluid into troughs, *reinforcing* the perturbation: exponential growth.



Variation of σ_n with n . Energy argument: KE $\sim (\rho_1 + \rho_2)v^2/(2k_n)$; PE released $\sim (\rho_2 - \rho_1)g\eta^2/2$. Equate, with $v \sim \sigma\eta$:

$$\sigma_n^2 \sim \frac{\rho_2 - \rho_1}{\rho_1 + \rho_2} g k_n.$$

Short waves (large n) localise motion at the interface where the baroclinic torque acts: faster growth. Finite depth h caps the inertia at $\sim h$ once $k_n h \lesssim 1$, suppressing σ_n by $\tanh(k_n h)$.

Question 4 — Stokes flow past a clean bubble

Problem. (a) Derive $\nabla^2 p, \nabla^2 \omega \approx 0$ in Stokes flow; (b) verify incompressibility of axisymmetric streamfunction and write BCs at the bubble; (c) verify $\Psi = Cr \sin^2 \theta$ and obtain drag $4\pi\eta a U$; (d) shrinking bubble at terminal velocity, find $a(z)$ and critical depth d for $a_0 = 10^{-4}$ m.

(a)

Low-Re conditions: $\text{Re} = UL/\nu \ll 1$; small U , small L , or large ν . Examples: bacterial swimming, microfluidics, lava/honey flow.

Drop inertia in Navier–Stokes:

$$0 \approx -\nabla p + \eta \nabla^2 \mathbf{u} + \rho \mathbf{g}.$$

Absorb gravity into modified pressure $p \rightarrow p + \rho g z$. Take divergence, use $\nabla \cdot \mathbf{u} = 0$:

$$\boxed{\nabla^2 p \approx 0.}$$

Take curl, use $\nabla \times \nabla p = 0$ and $\nabla \times \nabla^2 \mathbf{u} = \nabla^2 \boldsymbol{\omega}$:

$$\boxed{\nabla^2 \boldsymbol{\omega} \approx 0.}$$

(b)

Axisymmetric divergence:

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial(r^2 v_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sin \theta v_\theta)}{\partial \theta}.$$

Substitute streamfunction definitions:

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2 \sin \theta} \frac{\partial^2 \Psi}{\partial r \partial \theta} - \frac{1}{r^2 \sin \theta} \frac{\partial^2 \Psi}{\partial \theta \partial r}.$$

Schwarz cancellation:

$$\nabla \cdot \mathbf{v} = 0 \checkmark.$$

BCs at the bubble. Lab frame: far field at rest, bubble surface moves at $U \hat{\mathbf{z}}$. Kinematic match at $r = a$:

$$v_r(a, \theta) = U \cos \theta, \quad v_r \rightarrow 0, \quad v_\theta \rightarrow 0 \text{ as } r \rightarrow \infty.$$

Inviscid interior $\Rightarrow$ tangential viscous stress vanishes:

$$\sigma_{r\theta}|_{r=a} = 2\eta e_{r\theta}|_{r=a} = 0.$$

Use $v_r(a, \theta) = U \cos \theta \Rightarrow \partial v_r / \partial \theta|_a = -U \sin \theta$:

$$e_{r\theta}|_{r=a} = \frac{a}{2} \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) \Big|_{r=a} + \frac{1}{2a} (-U \sin \theta) = 0.$$

Rearrange the tangential stress condition:

$$\boxed{\frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) \Big|_{r=a} = \frac{U \sin \theta}{a^2}.$$

(c)

Given streamfunction (lab frame):

$$\Psi = Cr \sin^2 \theta.$$

Compute v_r :

$$v_r = \frac{1}{r^2 \sin \theta} \frac{\partial \Psi}{\partial \theta} = \frac{2C \cos \theta}{r}.$$

Compute v_θ :

$$v_\theta = -\frac{1}{r \sin \theta} \frac{\partial \Psi}{\partial r} = -\frac{C \sin \theta}{r}.$$

Apply $v_r(a, \theta) = U \cos \theta$:

$$\frac{2C \cos \theta}{a} = U \cos \theta.$$

Solve:

$$C = \frac{Ua}{2}.$$

Far-field check $r \rightarrow \infty$:

$$v_r \rightarrow 0, \quad v_\theta \rightarrow 0 \checkmark.$$

Form v_θ/r and differentiate:

$$\frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) = \frac{2C \sin \theta}{r^3}.$$

Evaluate at $r = a$ with $C = Ua/2$:

$$\left. \frac{\partial}{\partial r} \left(\frac{v_\theta}{r} \right) \right|_{r=a} = \frac{U \sin \theta}{a^2} \checkmark.$$

This matches the tangential-stress BC derived in part (b).

Drag. z -component of surface traction:

$$T_z = (-p + 2\eta e_{rr}) \cos \theta - 2\eta e_{r\theta} \sin \theta.$$

Tangential stress vanishes at $r = a$ (free surface):

$$2\eta e_{r\theta}|_{r=a} = 0.$$

Compute $\partial v_r / \partial r$ from $v_r = 2C \cos \theta / r$:

$$\frac{\partial v_r}{\partial r} = -\frac{2C \cos \theta}{r^2}.$$

Evaluate at $r = a$ with $C = Ua/2$:

$$\left. \frac{\partial v_r}{\partial r} \right|_{r=a} = -\frac{U \cos \theta}{a}.$$

Hence radial viscous stress at the surface:

$$2\eta e_{rr}|_{r=a} = -\frac{2\eta U \cos \theta}{a}.$$

Pressure on the surface (given form, $r = a$):

$$p|_{r=a} = p_0 + \frac{\eta U \cos \theta}{a}.$$

Combine $T_z = (-p + 2\eta e_{rr}) \cos \theta$ at $r = a$:

$$T_z|_{r=a} = -\left(p_0 + \frac{\eta U \cos \theta}{a} \right) \cos \theta - \frac{2\eta U \cos^2 \theta}{a}.$$

Collect terms:

$$T_z|_{r=a} = -p_0 \cos \theta - \frac{3\eta U \cos^2 \theta}{a}.$$

Net force from T_z over the sphere with $dA = a^2 \sin \theta d\theta d\phi$:

$$F = \int_0^{2\pi} \int_0^\pi T_z a^2 \sin \theta d\theta d\phi.$$

Do the ϕ integral:

$$F = 2\pi a^2 \int_0^\pi \left[-p_0 \cos \theta - \frac{3\eta U \cos^2 \theta}{a} \right] \sin \theta d\theta.$$

p_0 term vanishes since $\int_0^\pi \cos \theta \sin \theta \, d\theta = 0$:

$$F = -2\pi a^2 \cdot \frac{3\eta U}{a} \cdot \int_0^\pi \cos^2 \theta \sin \theta \, d\theta.$$

Evaluate $\int_0^\pi \cos^2 \theta \sin \theta \, d\theta = [-\cos^3 \theta / 3]_0^\pi = 2/3$:

$$F = -2\pi a^2 \cdot \frac{3\eta U}{a} \cdot \frac{2}{3}.$$

Simplify:

$$F = -4\pi\eta a U.$$

Magnitude:

$$\boxed{|F| = 4\pi\eta a U.}$$

(2/3 of Stokes drag on rigid sphere; tangential stress missing at the free surface.)

(d)

Terminal velocity: buoyancy = drag (with $\rho_a \ll \rho_w$):

$$\frac{4}{3}\pi a^3 \rho_w g = 4\pi\eta_w a U.$$

Solve for U :

$$U(a) = \frac{\rho_w g a^2}{3\eta_w}.$$

Bubble mass:

$$M = \frac{4}{3}\pi a^3 \rho_a.$$

Differentiate (with ρ_a constant):

$$\frac{dM}{dt} = 4\pi a^2 \rho_a \frac{da}{dt}.$$

Equate to dissolution rate $dM/dt = -4\pi a D_a \rho_a$:

$$4\pi a^2 \rho_a \frac{da}{dt} = -4\pi a D_a \rho_a.$$

Cancel:

$$a \frac{da}{dt} = -D_a.$$

Convert $dt = dz/U$ (bubble rising):

$$a U(a) \frac{da}{dz} = -D_a.$$

Substitute $U(a)$:

$$\frac{\rho_w g a^2}{3\eta_w} \cdot a \frac{da}{dz} = -D_a.$$

Simplify:

$$\frac{\rho_w g}{3\eta_w} a^3 \frac{da}{dz} = -D_a.$$

Separate and integrate from a_0 at $z = 0$:

$$\frac{\rho_w g}{3\eta_w} \int_{a_0}^a a'^3 \, da' = -D_a z.$$

Evaluate the integral:

$$\frac{\rho_w g}{3\eta_w} \cdot \frac{a^4 - a_0^4}{4} = -D_a z.$$

Multiply both sides by $12\eta_w/(\rho_w g)$:

$$a^4 - a_0^4 = -\frac{12\eta_w D_a}{\rho_w g} z.$$

Rearrange:

$$a^4(z) = a_0^4 - \frac{12\eta_w D_a}{\rho_w g} z.$$

Critical depth. Set $a(d) = 0$:

$$0 = a_0^4 - \frac{12\eta_w D_a}{\rho_w g} d.$$

Solve for d :

$$d = \frac{\rho_w g a_0^4}{12\eta_w D_a}.$$

Numerical values: $a_0 = 10^{-4} \text{ m}$, $\rho_w = 10^3 \text{ kg m}^{-3}$, $g = 9.81 \text{ m s}^{-2}$, $\eta_w = 10^{-3} \text{ Pa s}$, $D_a = 10^{-10} \text{ m}^2 \text{ s}^{-1}$.

Compute a_0^4 :

$$a_0^4 = (10^{-4})^4 = 10^{-16} \text{ m}^4.$$

Numerator:

$$\rho_w g a_0^4 = (10^3)(9.81)(10^{-16}) = 9.81 \times 10^{-13}.$$

Denominator:

$$12\eta_w D_a = 12(10^{-3})(10^{-10}) = 1.2 \times 10^{-12}.$$

Divide:

$$d = \frac{9.81 \times 10^{-13}}{1.2 \times 10^{-12}} \text{ m}.$$

$$d \approx 0.82 \text{ m}.$$